Problem 5

What do the events $A + A$ and AA represent?

Solution

We will interpret these operations using set theory notation.

Part 1: Event $A + A$ (Union)

Definition

$A + A$ represents the union of event A with itself:

$$A + A = A \cup A$$

Set-Theoretic Analysis

By the definition of union, $A \cup A$ consists of all elements that are in A or in A (or both).

For any element x:

$$x \in A \cup A \iff x \in A \text{ or } x \in A$$

Since "$x \in A$ or $x \in A$" is logically equivalent to "$x \in A$", we have:

$$A \cup A = A$$

Conclusion

$$\boxed{A + A = A}$$

Interpretation

The union of an event with itself is just the event itself. This is the **idempotent property of union**.

In probabilistic terms: "Event A occurs or event A occurs" is the same as "event A occurs."

Part 2: Event AA (Intersection)

Definition

AA represents the intersection of event A with itself:

$$AA = A \cap A$$

Set-Theoretic Analysis

By the definition of intersection, $A \cap A$ consists of all elements that are in A and in A.

For any element x:

$$x \in A \cap A \iff x \in A \text{ and } x \in A$$

Since "$x \in A$ and $x \in A$" is logically equivalent to "$x \in A$", we have:

$$A \cap A = A$$

Conclusion

$$\boxed{AA = A}$$

Interpretation

The intersection of an event with itself is just the event itself. This is the **idempotent property of intersection**.

In probabilistic terms: "Event A occurs and event A occurs" is the same as "event A occurs."

General Principle: Idempotent Laws

Both operations demonstrate the **idempotent laws** in set theory and Boolean algebra:

Operation	**Formula**	**Name**
Union	$A \cup A = A$	Idempotent law for union
Intersection	$A \cap A = A$	Idempotent law for intersection

Why "Idempotent"?

The term "idempotent" comes from Latin:

- *idem* = "same"
- *potent* = "power"

It means that applying the operation multiple times gives the same result as applying it once.

Verification

Using Venn Diagrams

- $A + A$: Shading region A or region A gives just region A
- AA: Shading region A and region A gives just region A

Using Element Membership

For any element x:

- $x \in A + A \iff (x \in A) \vee (x \in A) \iff x \in A$
- $x \in AA \iff (x \in A) \wedge (x \in A) \iff x \in A$

Both simplify to $x \in A$, confirming that $A + A = A$ and $AA = A$.

Examples

Example 1: Die Roll

Let $A =$ "rolling an even number on a die" $= \{2, 4, 6\}$.

Then:

- $A + A =$ "rolling an even number or rolling an even number" $= \{2, 4, 6\} = A$
- $AA =$ "rolling an even number and rolling an even number" $= \{2, 4, 6\} = A$

Example 2: Card Draw

Let $A =$ "drawing a red card" from a standard deck.

Then:

- $A + A =$ "drawing a red card or drawing a red card" $=$ "drawing a red card" $= A$
- $AA =$ "drawing a red card and drawing a red card" $=$ "drawing a red card" $= A$

Contrast with Probability

Note that while $A + A = A$ and $AA = A$ as events (sets), their probabilities behave differently:

- $P(A + A) = P(A)$ (correct)
- $P(AA) = P(A)$ only if we interpret this as a single event
- But if A occurred on two independent trials, then $P(A \text{ on trial 1 and } A \text{ on trial 2}) = P(A) \cdot P(A) = [P(A)]^2$

The notation AA in the context of a single trial means the same event A, so $P(AA) = P(A)$.

Final Answer

$$\boxed{A + A = A \quad \text{(Union of an event with itself)}}$$

$$\boxed{AA = A \quad \text{(Intersection of an event with itself)}}$$

Interpretation:

- $A + A = A$: "Event A occurs or event A occurs" is the same as "event A occurs"
- $AA = A$: "Event A occurs and event A occurs" is the same as "event A occurs"

General Principle: These are the **idempotent laws** of set operations. An event combined with itself (by union or intersection) remains unchanged.

Key Properties:

- Both are examples of idempotence: $A \circ A = A$
- This holds for any set/event A
- In Boolean algebra: $x \vee x = x$ and $x \wedge x = x$